

MUHSIN KALODI

Machine Learning Engineer | Full-Stack Software Developer

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Professional Summary

Machine Learning Engineer and Full-Stack Software Developer with hands-on experience building machine learning models, AI-powered applications, and scalable web systems. Proficient in Python, TensorFlow, Scikit-learn, Next.js, Node.js, PostgreSQL, and Docker. Passionate about developing production-ready AI solutions, enterprise software, and data-driven applications using modern software engineering practices.

Technical Skills

Languages: Python, JavaScript (ES6+), SQL, Java

Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, CNNs, Feature Engineering, Model Training, Model Evaluation

Web Development: Next.js, React.js, Node.js, Express.js, Flask, Django. REST APIs, HTML5, CSS3

Databases: PostgreSQL, MongoDB Atlas, SQLite, Firebase

Tools: Git, GitHub, Docker, Linux, Postman, Jupyter Notebook, Power BI

Professional Experience

Machine Learning Engineer Intern

Dec 2024 – Jan 2025

WexorAI, Kozhikode, Kerala

- Developed end-to-end machine learning pipelines involving data preprocessing, feature engineering, model training, validation, and inference using Python and Scikit-learn.
- Improved model performance through feature engineering, hyperparameter tuning, and cross-validation while evaluating models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Automated data preprocessing workflows using Pandas and NumPy, improving data quality, consistency, and reducing manual data preparation efforts.
- Collaborated with software engineers to integrate machine learning solutions into scalable applications using modular development practices and Git-based version control.

Selected Projects

Student Performance Prediction Analysis

Python | Scikit-learn | Pandas | NumPy | Matplotlib

- Developed a machine learning system to predict student academic performance using historical academic and behavioural datasets.
- Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and trained multiple supervised learning models to identify the best-performing solution.
- Evaluated models using Accuracy, Precision, Recall, F1-Score, MAE, and RMSE while visualizing insights to support data-driven academic decision making.

StudentDex – Student Information Management System

Next.js | Node.js | PostgreSQL | Docker | REST APIs

- Engineered a full-stack student management platform for managing academic records, attendance, student profiles, and administrative operations.
- Developed secure REST APIs with PostgreSQL integration, role-based authentication, and automated Excel-to-database import functionality.
- Applied modular architecture, Docker containerization, and Git version control to improve maintainability, scalability, and deployment consistency.

CHEDI – AI Plant Disease Detection

Python | TensorFlow | OpenCV | Streamlit

- Built an AI-powered computer vision application for plant disease detection using Convolutional Neural Networks (CNNs).
- Implemented image preprocessing, augmentation, and TensorFlow-based model training to improve classification performance.
- Developed an interactive Streamlit application for real-time plant disease prediction from uploaded leaf images.

Education

Bachelor of Technology in Artificial Intelligence and Data Science	2022 – 2026
Excel Engineering College, Komarapalayam, Namakkal, Tamil Nadu	CGPA: 7.0/10

Certifications

- Google AI Essentials – Coursera (2024)
- Google Foundations of Cybersecurity – Coursera (2024)
- Play It Safe: Manage Security Risks – Coursera (2024)
- Cisco: Introduction to Modern AI (2025)
- Cisco: Introduction to Cybersecurity (2025)
- NPTEL SWAYAM: Introduction to Industrial Internet of Things (Jan–Apr 2025)
- GUVI: Mastering MongoDB (2025)
- A Deep Dive into MongoDB Atlas for Modern Web Development – NIT Trichy Pragyan Workshop (2024)

Leadership & Activities

- National Service Scheme (NSS) Volunteer (2022)
- Google Developer Groups (GDG) Community Participant
- FOSSMeet Open-Source Community Participant
- Participated in TECBLAZE 2024 at SRM Institute of Science and Technology, Ramapuram, including Workshop, WEBZEN, and Quizzeria technical events.